

Python for Data Science

Sample knowledge-base document for a Retrieval-Augmented Generation project.

Python Basics

Python is a high-level programming language widely used in data science and machine learning. Variables do not require explicit type declarations. Common built-in types include integers, floats, strings, lists, tuples, dictionaries, and sets.

NumPy

NumPy provides efficient multidimensional arrays and numerical operations. It is commonly used for vectorized mathematics, matrix operations, random sampling, and numerical preprocessing.

Pandas

Pandas provides Series and DataFrame structures for tabular data. Common operations include loading CSV files, filtering rows, selecting columns, grouping values, joining tables, handling missing data, and computing summary statistics.

Scikit-learn

Scikit-learn provides machine learning algorithms and utilities for preprocessing, model selection, pipelines, evaluation metrics, and dataset splitting. A common workflow is to split data, fit a model, predict on test data, and evaluate the predictions.

Sentence Transformers

Sentence Transformers are neural models designed to create useful embeddings for sentences and paragraphs. In a RAG application, they can encode document chunks and user queries into vectors for semantic retrieval.

ChromaDB

ChromaDB is a vector database that can store document text, metadata, identifiers, and embeddings. It can query a collection using an embedding and return the nearest documents.

Good Project Structure

A small RAG project can use a data folder for source documents, a notebook or src folder for code, a vector database folder for persistent embeddings, and a requirements or pyproject file for dependencies.